Face Verification / Re-Identification

ResNet50 embeddings, ArcFace + triplet training, cosine-similarity verification and gallery/probe identification

1. Summary

A ResNet50 backbone fine-tuned with a combined ArcFace + batch-hard-triplet objective produces 512-D L2-normalised face embeddings. Evaluated on **120 identities never seen during training**, cosine similarity gives **ROC-AUC = 0.9700**, **EER = 8.98%** and verification accuracy **90.94%** at a threshold of **0.1792**. Closed-set identification reaches **Rank-1 = 55.87%** and **Rank-5 = 81.89%**.

2. Dataset

Labeled Faces in the Wild (funneled)
Canonical source: http://vis-www.cs.umass.edu/lfw/ (offline since Aug 2026)
Downloaded from: https://ndownloader.figshare.com/files/5976015
Licence: Free for non-commercial research use; images from public news photographs. Images come from public news photographs; no law-enforcement or criminal-record imagery is involved.

Property	Value
Identities	1680
Images	9164
Resolution	224 x 224 RGB
Train	1500 identities / 7096 images
Validation	60 identities / 1037 images
Test	120 identities / 1031 images
Split policy	Identity-disjoint
Face detection	OpenCV Haar cascade (frontalface_default)
Alignment	eye-line rotation via haarcascade_eye when both eyes found

Splits are drawn over **identities**, not images. Every person used for validation or testing is absent from training, so the reported numbers measure generalisation to unseen people.

3. Model and training strategy

Architecture. Face image (3x224x224) -> ResNet50 trunk (ImageNet-pretrained, classifier removed) -> global average pooling (2048-D) -> Dropout -> Linear(512, no bias) -> BatchNorm1d -> L2 normalisation -> 512-D embedding on the unit hypersphere.

Objective. ArcFace additive-angular-margin cross-entropy over the training identities plus a batch-hard triplet term on the same embeddings. ArcFace was chosen over plain softmax because the system scores faces by cosine similarity: normalising embeddings and class weights makes training optimise the angular geometry that verification actually measures, and the margin forces an explicit gap between identities. The triplet term is added because the classifier head is discarded at inference - it organises samples around learned class centres, whereas batch-hard mining works directly on the sample-to-sample cosine distances the ROC is built from. Batches are drawn with a P x K sampler so every batch has valid positives to mine.

Transfer learning. With only a few thousand training images a ResNet50 trained from scratch overfits badly, so the trunk starts from ImageNet weights. A linear warm-up followed by cosine decay protects them early on, and the randomly-initialised embedding and ArcFace heads use a 10x larger learning rate than the trunk.

Training detail	Value

Backbone	resnet50
Training set	7096 images / 1500 identities
Epochs	30
Best epoch (validation AUC)	22
Validation AUC	0.9636
Validation EER	9.68%
Train accuracy at best epoch	82.54%
Loss: first -> last epoch	20.28 -> 2.01
Time per epoch	41 s (Colab T4)
Selection criterion	Highest verification AUC on validation identities

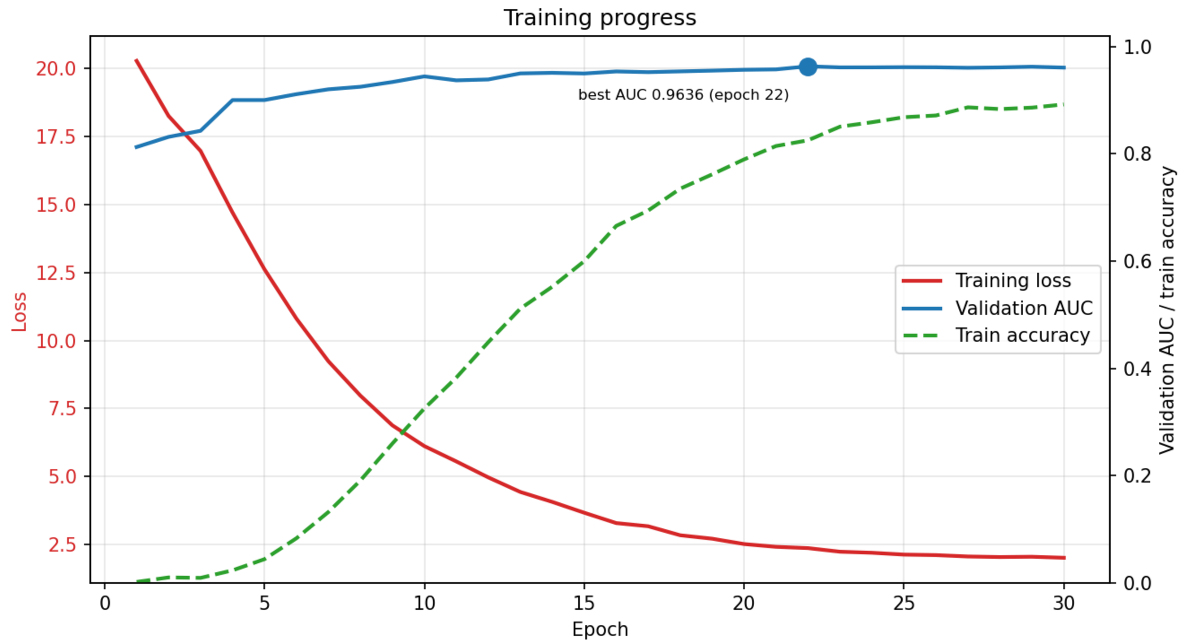

Figure 1 - Training loss against validation AUC. The gap between rising train accuracy and a flattening validation AUC after ~epoch 20 is the point where the model starts fitting the training identities rather than learning transferable features.

4. Pair generation

Property	Value
Split	test
Positive pairs	5000
Negative pairs	5000
Identities	120
Images	1031
Max possible positives	7861
Duplicate pairs	0
Seed	42

Positive pairs are sampled **round-robin across identities** rather than uniformly over all within-identity combinations. LFW is severely imbalanced - a single identity can contribute tens of thousands of the possible genuine pairs - and uniform sampling would let a handful of people dominate the genuine score distribution, producing an optimistic and unrepresentative ROC. Negative pairs draw two distinct identities first, then one image from each. Every pair is keyed by its sorted path tuple in a hash set, so (A,B) and (B,A) collapse to one entry and no pair repeats; an image is never paired with itself. Pairs are drawn exclusively from the identity-disjoint test split and the model plays no part in selecting them, so neither training data nor model bias can leak into the evaluation.

5. Verification results

Metric	Value
ROC-AUC	0.9700
EER	8.98%
Selected threshold	0.1792
Threshold selected on	validation split (pair_scores_val.csv)
Accuracy at threshold	90.94%
Precision	0.9186
Recall	0.8984
F1 score	0.9084
TAR @ FAR = 1%	67.66%
TAR @ FAR = 0.1%	37.60%
False accepts	398 / 5000
False rejects	508 / 5000
Genuine mean +/- std	0.3984 +/- 0.1751
Impostor mean +/- std	0.0125 +/- 0.1144

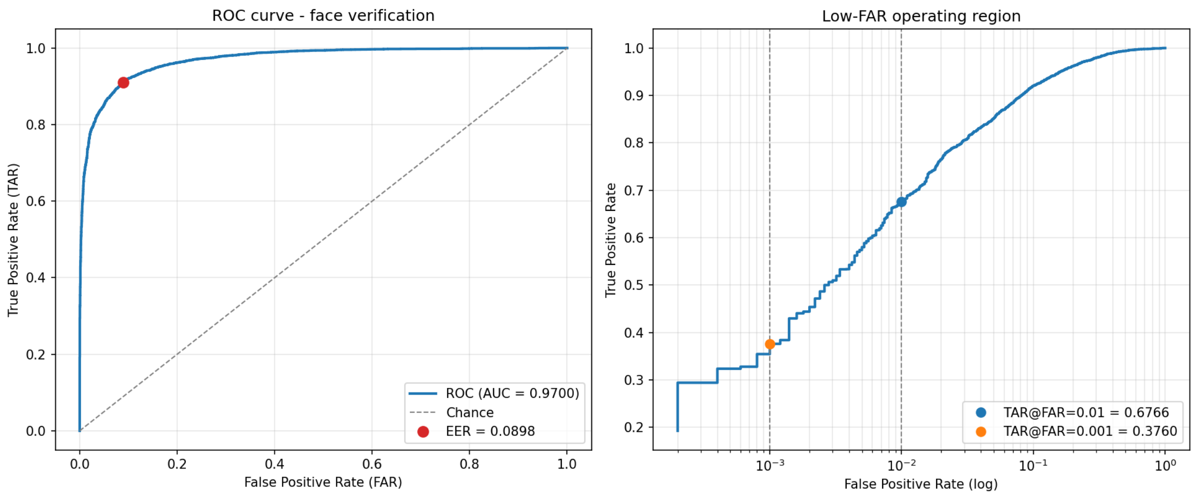

Figure 2 - ROC curve (linear) and the low-FAR operating region (log FAR).

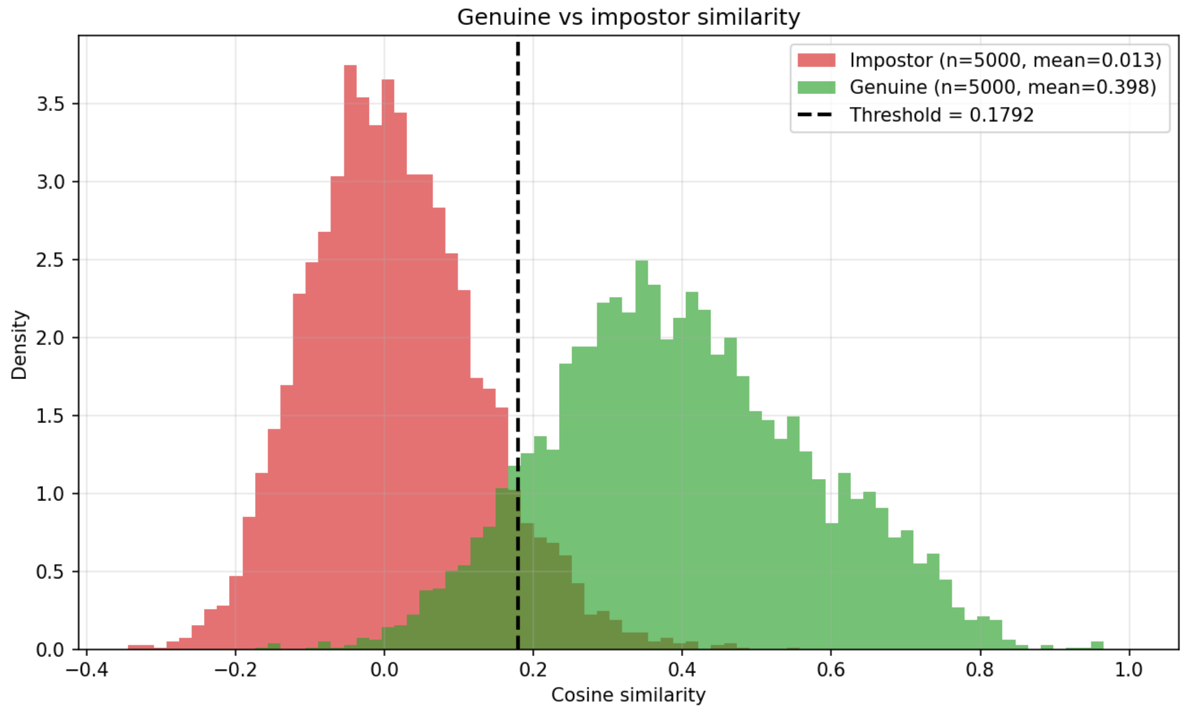

Figure 3 - Genuine and impostor cosine-similarity distributions with the decision threshold. The overlap is where verification errors occur.

6. Threshold analysis

Decision rule: cosine similarity >= 0.1792 -> MATCH, otherwise NON-MATCH. The threshold maximises verification accuracy on the validation split and is then applied unchanged to the test split. Selecting it on the test scores would be a mild form of leakage and would overstate deployed performance.

Raising the threshold makes the system more suspicious: fewer pairs clear the bar, so false accepts fall while false rejects rise - the right operating point for access control, where wrongly admitting a stranger costs far more than asking a legitimate user to retry. Lowering it does the reverse, trading security for convenience. The two error rates cannot both be reduced at a fixed embedding; only a better embedding lifts the whole ROC curve.

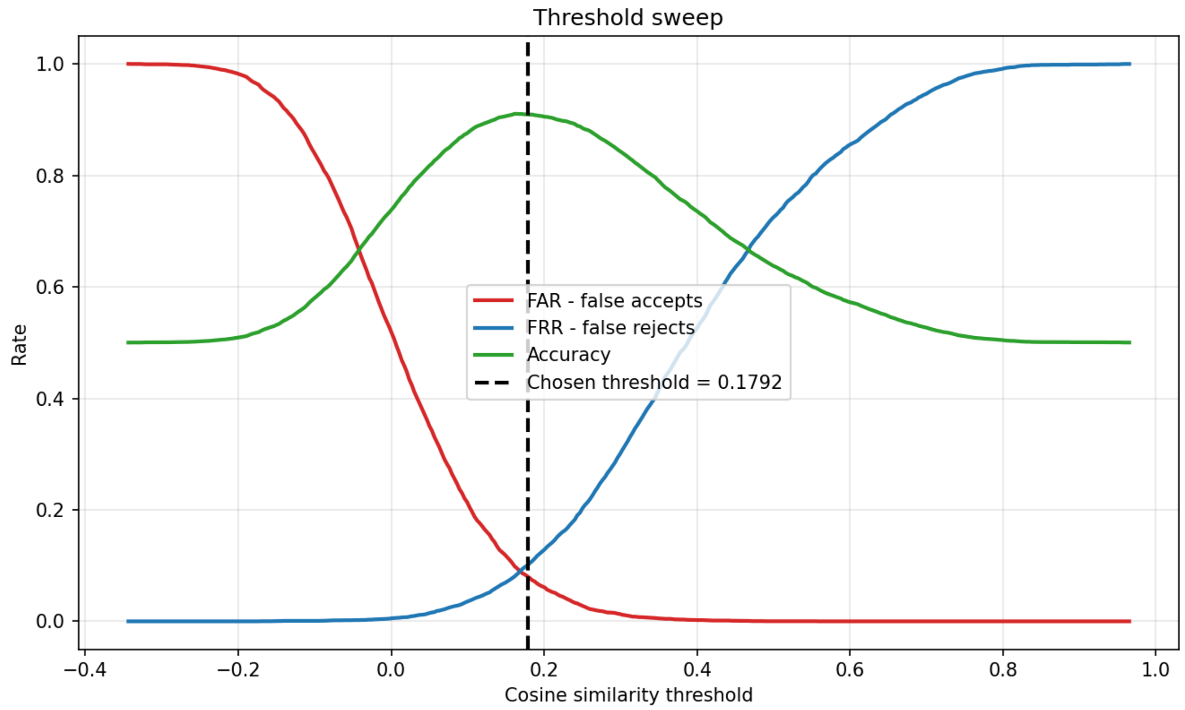

Figure 4 - FAR, FRR and accuracy across the threshold range. The FAR/FRR crossing point is the EER.

Resolving FAR below 1e-4. The 5,000 + 5,000 sampled evaluation set cannot resolve a false-accept rate smaller than 1/5000 = 2e-4, so its low-FAR operating points rest on a handful of pairs. Scoring every pair in the test split instead - all C(1031, 2) of them - raises the impostor count from 5,000 to 523,104 and makes FAR down to 1.9e-06 measurable.

Operating point	TAR (impostor pairs at that FAR)
FAR = 0.01	67.07% (5231 pairs)
FAR = 0.001	40.34% (523 pairs)
FAR = 0.0001	23.14% (52 pairs)
FAR = 1e-05	12.64% (5 pairs)
ROC-AUC (all pairs)	0.9710
EER (all pairs)	8.92%

AUC and EER agree with the sampled protocol to within 0.001 and 0.06 points, confirming the round-robin pair sampling did not bias the score distribution. What changes is resolution: TAR@FAR=1e-3 moves from 37.60% to 40.34%, and the sampled figure was the noisy one - roughly five impostor pairs against 523. The headline metrics in this report stay on the 5k/5k protocol the assessment specifies; the exhaustive run is reported alongside because low-FAR claims need the extra pairs to mean anything.

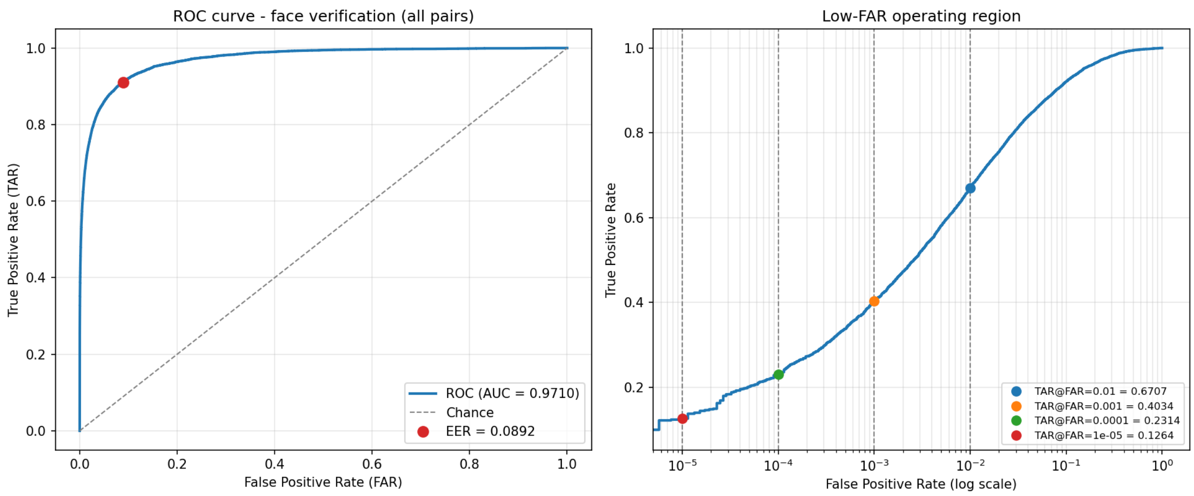

Figure 5 - ROC over all 530,965 test pairs. The log-FAR panel now extends to 1e-5, where the curve is smooth rather than the staircase the sampled set produces below 1e-3.

Attempts to improve low-FAR performance. TAR@FAR=0.1% is the weakest reported metric, so two standard families of post-processing were tried, each fitted on the training split only and selected on validation before being measured on test. Embedding transforms (mean-centering, ZCA whitening) and cohort score normalisation (Z-norm, S-norm, adaptive S-norm) were both rejected. Whitening gained 3.8 points of TAR@FAR=0.1% on validation and lost 9.8 on test; adaptive S-norm gained 3.2 on validation and lost 1.7 on test. Three validation gains in a row failed to replicate, which is exactly what should be expected when FAR = 0.1% corresponds to about five impostor pairs. Critically, none of these transforms moved AUC at all - when post-processing cannot extract more signal, the constraint is the embedding rather than the scoring rule, which points back at the 7,096-image training set. Full tables are in the README.

7. Gallery / probe identification

Each of the 120 test identities is enrolled with its first 1 image(s); the remaining 911 images act as probes. A probe embedding is compared against every gallery template by cosine similarity and assigned the highest-scoring identity.

Metric	Value
Gallery identities	120
Gallery images	120
Probe images	911
Rank-1 accuracy	55.87%
Rank-5 accuracy	81.89%
Chance Rank-1	0.83%
Mean rank of true identity	4.51

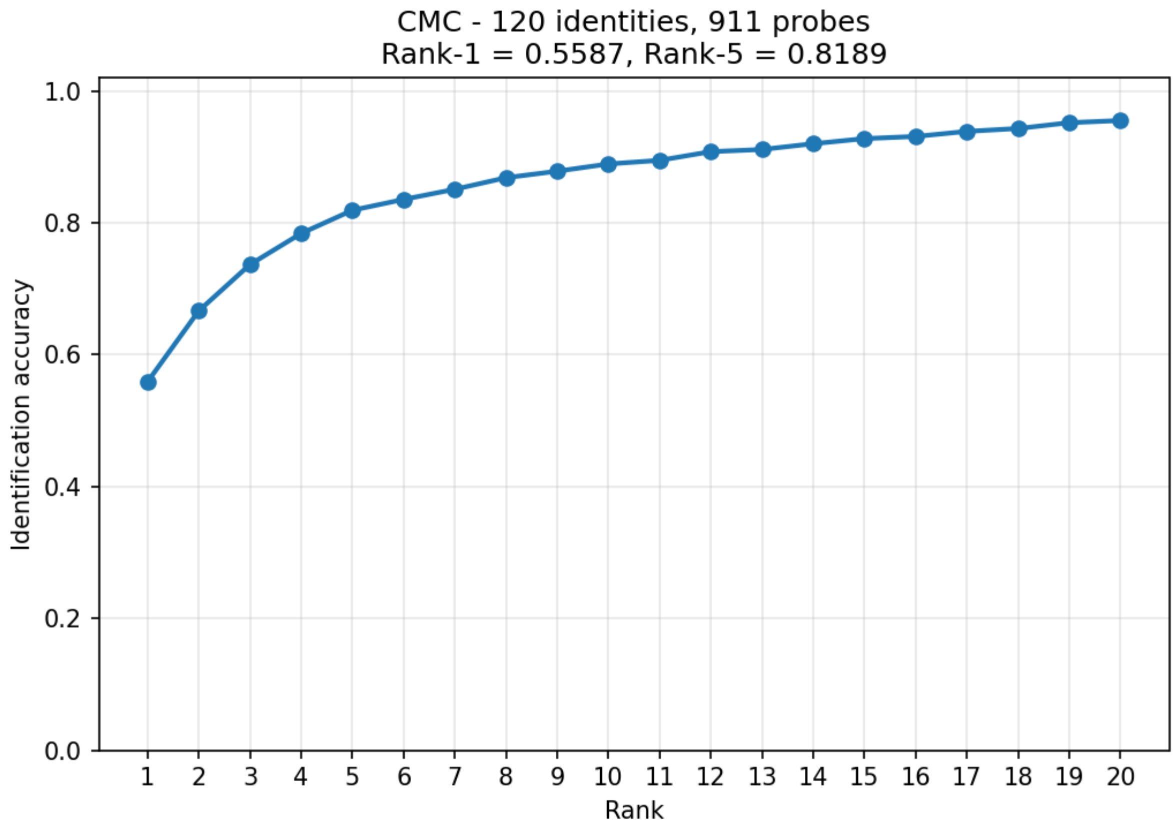

Figure 6 - CMC curve: probability the correct identity is within the top-k gallery matches.

Single-image enrolment is the hardest setting. Enrolling more images per identity averages several embeddings into each template, which cancels pose and lighting noise and raises accuracy sharply - at the cost of fewer probes left to test with.

Images enrolled per identity	Rank-1 / Rank-5 / mean rank / probes
1	55.87% / 81.89% / 4.51 / 911
2	72.82% / 91.28% / 2.55 / 791
3	75.86% / 94.19% / 2.13 / 671

8. Open-set identification

Closed-set Rank-1 assumes every probe belongs to someone in the gallery, so the system can always answer. A deployed system also has to be able to say "I don't know". Here 40 of the test identities are withheld from the gallery entirely and all 399 of their images become impostor probes, so accepting one is a false alarm rather than a ranking mistake. A probe is only counted correct if its top-1 similarity clears the threshold *and* points at the right person.

Metric	Value
Enrolled identities	80
Unknown identities	40
Known / unknown probes	552 / 399
Closed-set Rank-1 (same gallery)	64.13%
DIR @ FPIR = 1%	20.11%
Threshold @ FPIR = 1%	0.4852

DIR @ FPIR = 10%	42.75%
Threshold @ FPIR = 10%	0.3947
Mean top-1 similarity, known	0.4107
Mean top-1 similarity, unknown	0.3175

Performance drops sharply: 64.13% closed-set becomes 20.11% once the system must also reject strangers at a 1% false-alarm rate. The reason is visible in the last two rows - the mean top-1 score for an unknown face (0.3175) is not far below that of a genuine match (0.4107), because the nearest of 80 enrolled faces is often a plausible look-alike. Separating those needs a stronger embedding, not a better threshold.

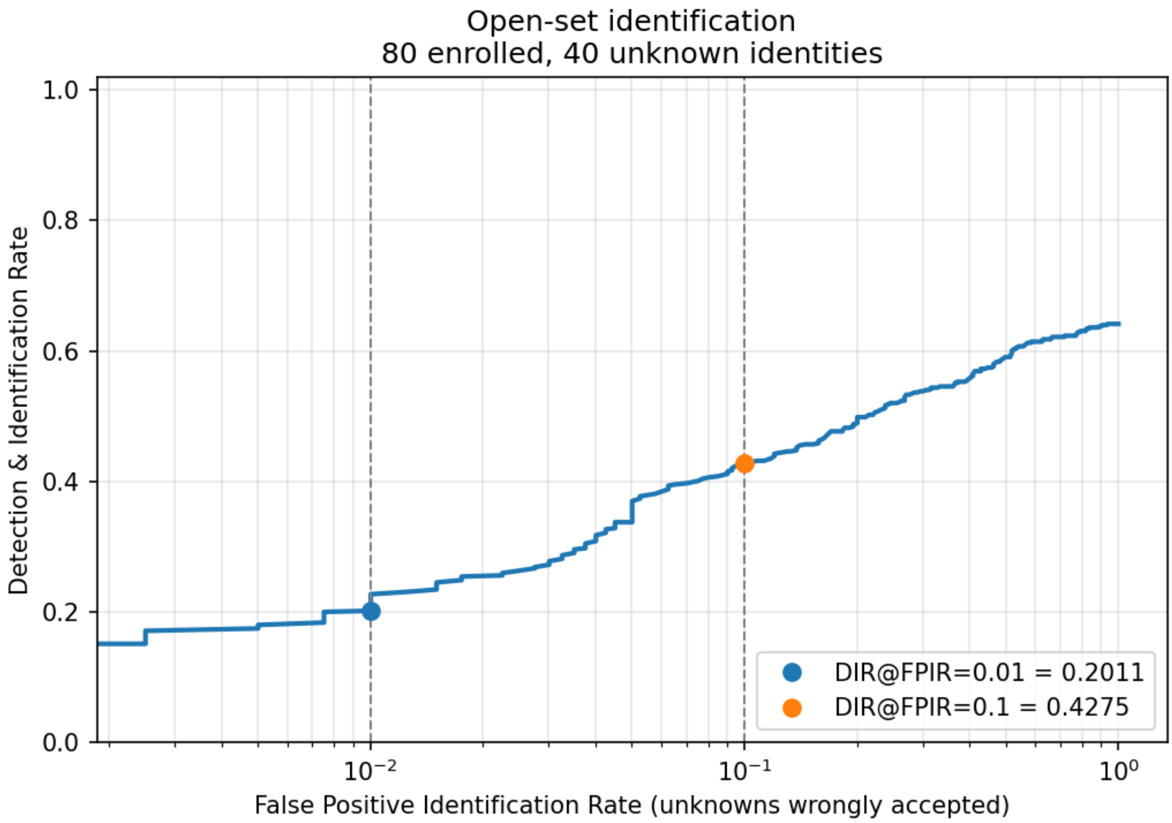

Figure 7 - Detection & identification rate against the rate at which unknown people are wrongly accepted.

9. Known limitations

Dataset scale. A few thousand training images across ~1.5k identities is two to three orders of magnitude smaller than the corpora (MS1M, WebFace260M) behind production face recognition, so absolute accuracy sits below published state-of-the-art LFW figures, which train on large external datasets.

Demographic bias. LFW comes from 2000s news photography and skews heavily towards light-skinned adult males in frontal, well-lit poses. Error rates on under-represented groups will be materially worse than the aggregate numbers here, which are not evidence of fairness across demographics.

Detection and alignment. The Haar cascade is fast and dependency-free but misses profile and heavily-occluded faces; those images fall back to a fixed central crop, safe on funnelled LFW imagery but unreliable in the wild. A modern detector with five-point landmark alignment (RetinaFace, MTCNN, YuNet) would improve crop consistency and typically several points of accuracy.

Open-set performance is weak. Section 8 measures it rather than assuming it away, and the answer is that rejecting strangers at a 1% false-alarm rate costs roughly three quarters of the closed-set accuracy. That is the number a deployment would actually live with, and it is the clearest evidence that the embedding needs more training data before it could be used for anything real.

Single split, single seed. Metrics come from one identity split and one training run; confidence intervals would need k-fold splits or repeated seeds.

No liveness detection. The system compares images, not live subjects - a printed photograph would verify as readily as a real face.

Not a benchmark comparison. These numbers use a custom identity-disjoint split, not the official LFW 6,000-pair protocol, so they are not directly comparable to published LFW accuracies.